

Wenrui XU

Room 412, Bauer Wurster Hall, University of California, Berkeley, CA 94720
wenruixu@outlook.com | +1(510)229-8310 | +86 139-0569-6236

EDUCATION

Ph.D. Student, Environmental Planning, Dept. of Landscape Arch. & Envi. Planning 2025-2029(Expected)
UNIVERSITY OF CALIFORNIA, BERKELEY CA, USA

- *Advisor*: Prof. Lu Liang, Geospatial 3M Lab (Monitoring-Mapping-Modeling)
- *Research Interests*: Geographic Information Science, Remote Sensing, Compounding Environmental Risks, Geospatial Data Science & AI, Urban Planning and Design, Machine Learning, Multimodal LLM

Master of Architecture, Urban Informatics Program, School of Architecture 2023-2025
TSINGHUA UNIVERSITY Beijing, China

- *GPA*: 3.93/4.00 | *GRE*: 336 (Verbal 166, Quantitative 170) | *TOEFL*: 107 (Reading 30, Listening 29, Writing 26)
- *Honor*: Comprehensive Excellence Scholarship
- *Main Courses*: Big Data & Urban Planning (A+); Urban Informatics (A-); Urban Renewal & Governance (A)
- *Certificate in Global Private Equity Research Institute*: Corporate Finance (A); Analyzing Financial Statements(A-); Private Equity and Venture Capital (A); Institutional Buy-side Investment in China (A-)
- *Graduation Project*: Planning and Design of Zhejiang Wanli University Yuyao Campus

Bachelor of Engineering, Architecture and Urban Design, College of Arch. & Urban Planning 2019-2023
TONGJI UNIVERSITY Shanghai, China

- *GPA*: 4.89/5.00 | *Ranking*: 1% | *CET-6*: 684 | *CET-4*: 642
- *Honor*: Graduation with the Highest Distinction (Top 1% of all graduates in Tongji University and Shanghai) National Scholarship (Top 1%); Outstanding Student model (7/4300+ undergraduates) Outstanding Student Leader; The First Prize Scholarships; Enterprise Scholarships
- *Graduation Project*: Port-Oriented and Landscape Urbanism: Hong Kong-Shenzhen Border Urban Design

FUDAN UNIVERSITY

- *Minor Program*: Macroeconomics (A); Microeconomics (A-); Accounting (A); Financial Market (A)

TEACHING EXPERIENCE

UNIVERSITY OF CALIFORNIA, BERKELEY (First-Time GSI Prerequisites Fulfilled)

- [Fall 2025] ESPM 375 Teaching in Environmental Science, Policy, and Management
- [Fall 2025] Teaching Conference for First-Time GSIs
- [Fall 2025] GSI Professional Standards and Ethics Online Course

TSINGHUA UNIVERSITY

- [Fall 2023, Spring 2024, Fall 2024, Spring 2025] Undergraduate Thesis Studio (Teaching Assistant).
- [Fall 2023, Fall 2024] Urban Design Elements (Teaching Assistant).

TONGJI UNIVERSITY

- [Fall 2022] Building Mechanics and Structure (Tutor).

PROFESSIONAL EXPERIENCE

Research Assistant, FIB-Lab (Future Intelligence Lab), Dept. of Electronic Engineering 2024.09-2025.06
TSINGHUA UNIVERSITY Beijing, China

- Led a project on LLM Spatial Intelligence, published on ACL Main 2025 (CCF-A).
- *Advisor*: Prof. Chen Gao, Prof. Jie Feng, Prof. Yong Li

Intern Planner & Architect	2024.10-2025.03
ARCHITECTURAL DESIGN & RESEARCH INSTITUTE OF TSINGHUA UNIVERSITY (THAD)	Beijing, China
• Zhejiang Wanli University Yuyao Campus Project (Constructed): Landscape planning and architectural design for two college buildings	
• <i>Advisor:</i> Chief Architect Xiuquan Mo, Prof. Jinxi Chen	
Intern Architect	2023.08-2024.06
TSINGHUA URBAN PLANNING & DESIGN INSTITUTE (THUPDI)	Beijing, China
• Beijing 1st Experimental Primary School Honglian Branch Renewal Project (Constructed)	
• <i>Advisor:</i> Chief Architect Longlin Jin, Prof. Jinxi Chen	
Intern Equity Researcher	2023.08-2024.06
KAIYUAN SECURITIES RESEARCH INSTITUTE	Shanghai, China
• Published an in-depth Report on the cosmetics industry and BeiTaiNi (SZ.300957)	

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES

- [1] **Xu, W.**[†], Lyu, D.[†], Wang, W.[†], Feng, J., Gao, C., & Li, Y.* (2025). Defining and evaluating visual language models' basic spatial abilities: A perspective from psychometrics. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (volume 1: Long Papers) (pp. 11571–11590). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.acl-long.567>
- [2] **Xu, W.**, & Gu, X.* (2025). Spatiotemporal impacts of purpose-specific human mobility on air pollution: Evidence from taxi trajectories and interpretable machine learning. Sustainable Cities and Society, 126, 106411. <https://doi.org/10.1016/j.scs.2025.106411>
- [3] Gu, X., **Xu, W.**, Gong, C., & Liu, X.* (2025). City centers really lived up to the hype? Evidence from human perceptions of over 4000 communities in China. Cities, 166, 106278. <https://doi.org/10.1016/j.cities.2025.106278>
- [4] Wang, S., Zhao, Z., Xie, Y., Ma, M., Chen, Z., Wang, Z., Su, B., **Xu, W.**, & Li, T. (2024). Recent surge in public interest in transportation: Sentiment analysis of baidu apollo go using weibo data. arXiv. <https://doi.org/10.48550/arXiv.2408.10088>

WORKING PAPERS

- [1] Lyu, D.[†], **Xu, W.**[†], Wang, W., Gao, C., Zhuang, W., Li, Y. Aesthetic Perception of Large Language Models. Manuscript in Preparation.
- [2] Gu, X., Liu, Y., **Xu, W.**, Qiu, W., Li, X. Redefining thermal injustice with building façade materials across five US cities. Manuscript in Preparation.
- [3] Lyu, D., Li, J., **Xu, W.** Spatial Inequities in Cardiovascular Emergency Rescue Networks in Chinese Cities. Manuscript in Preparation.
- [4] **Xu, W.**, Liang, L. Urban Heat Mosaic: A Case Study in San Francisco. Manuscript in Preparation.
- [5] **Xu, W.**, Liang, L. Compound Heatwave-Air Pollution Risks in the United States. Manuscript in Preparation.
- [6] **Xu, W.**, Zhang, Z., Rui J. Data-Driven Analysis of Elevator Installation in Aging Residential Buildings: Spatial Equity, Influencing Factors, and Predictive Modeling in Beijing. Manuscript in Preparation.

PRESENTATIONS

- [2025] The 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Oral Presentation, Vienna, Austria.
- [2025] The 65th Association of Collegiate Schools of Planning Annual Conference (ACSP), Minneapolis, USA.
- [2025] The 4th International Conference on Urban Informatics (GSCS & ICUI), Oral Presentation, Hong Kong, China.
- [2025] The 19th International Conference on Computational Urban Planning and Urban Management (CUPUM), Poster Presentation, London, UK.

- [2025] The 19th International Association for China Planning Annual Conference (IACP), Oral Presentation, Xiamen, China.
- [2025] The 32nd International Conference on Geoinformatics (CPGIS), Oral Presentation, Jiaozuo, China.
- [2025] The 85th Annual Meeting of the Academy of Management (AOM), Extending Knowledge-based View in Generative AI Era, Copenhagen, Denmark.
- [2025] Strategic Management Society 45th Annual Conference (SMS), Architecting Knowledge Ecosystems: How Generative AI Redraws the Boundaries of Competitive Advantage, San Francisco, USA.
- [2025] The 25th COTA International Conference of Transportation Professionals, Guangzhou, China.
- [2024] Nature Conference on Air Pollution and Climate Change, Poster Presentation, Beijing, China.

AWARDS AND HONORS

- [2024] Comprehensive Excellence Scholarship: Awarded by Tsinghua University.
- [2023] Distinct Graduate of Shanghai: Graduation with the highest distinction (Top 1%).
- [2023] Jundi Travel Grant (~\$30,000)
- [2022] The First Prize Undergraduate Scholarship: Awarded by Tongji University.
- [2022] Exhibition of Architectural Design in Developing Countries: Bronze Award.
- [2021] The First Prize Undergraduate Scholarship: Awarded by Tongji University.
- [2021] Outstanding Student Model: Top 7 of 4300+ undergraduates at Tongji University (Top 0.1%).
- [2021] National Real Estate Innovation & Entrepreneurship Competition: Top Prize.
- [2021] National Computer Design Competition for College Students: Second Prize.
- [2020] National Scholarship: Highest honor for undergraduate (Top 1%).
- [2020] National English Competition for College Students: Top Prize.

PROFESSIONAL SERVICES

PEER REVIEWER

- GIScience & Remote Sensing; Building and Environment; Computational Urban Science; Architectural Engineering and Design Management; ICLR 2025 EmbodiedAI Workshop

CONFERENCES & WORKSHOPS

- [2024] Milan Summer Youth Festival, China-Italy Design and Innovation Hub, Tsinghua University, Student Volunteer Leader as “Interdisciplinary Design Innovation Scholars”.
- [2023, 2024] The 3rd / 4th International Symposium of Urban Design Research and Education, Tsinghua University, Conference Assistant.

VOLUNTEER

- [2023- Now] Citipedia (#1 volunteer group promoting sustainable city & transportation in CN), Volunteer.
- [2021-2022] Student Union of College of Arch. & Urban Planning, Tongji University, President.
- [2021-2022] The “Dream Classroom” voluntary teaching + design + construction project, Tibet Lhatse Middle School, Xinjiang Huocheng Middle School, Tongji University, Leader.
- [2021] Archdaily (International Architecture Media), Volunteer Chinese Translator & Editor

PROFESSIONAL AFFILIATIONS

- [2022-2024] Architectural Society of China-Student Branch (ASC-SAC), Committee Member.
- [2022- Now] Architectural Society of China (ASC), Student Member.
- [2025- Now] American Association of Geographers (AAG), Member.
- [2025- Now] Association of Collegiate Schools of Planning (ACSP), Member.
- [2024- Now] International Association for China Planning (IACP), Member.
- [2024- Now] International Association of Chinese Professionals in GIS (CPGIS), Member.

CERTIFICATES & SKILLS

- *Geospatial Data*: ArcGIS; QGIS; GEE; SPSS; MATLAB; Origin; Tableau
- *Machine Learning*: PyTorch; TensorFlow; CNN, FCN, GNN; XGBoost; SHAP
- *Urban Planning & Design*: Revit; Dynamo; Rhino; Grasshopper; SketchUp; CAD; V-Ray; Lumion
- *Graphics & Video*: Adobe Creative Suite (Photoshop, Illustrator, InDesign, AfterEffect, Premiere); DaVinci
- *Other*: HTML; CSS; Web Crawler; LaTeX; Git
- *Language*: English (Fluent); Chinese Mandarin (Native); Cantonese (Basic); Japanese (Basic)